

Module 2

Accounting Statements and Finance

FINE 3010, Financial Management
Prof. Renping Li, Tulane University

Why Do We Need Accounting?

1. What do firms own?
 - Real assets (tangible, intangible) and financial assets (cash, securities)

Why Do We Need Accounting?

1. What do firms own?
 - Real assets (tangible, intangible) and financial assets (cash, securities)
2. What do firms owe?

Why Do We Need Accounting?

1. What do firms own?
 - Real assets (tangible, intangible) and financial assets (cash, securities)
2. What do firms owe?
3. How much money did firms make?
 - Income? Cash flow?

Example, Is Anthropic Profitable?

- Is Anthropic profitable in Q2 2026?



CNBC, August 18, 2026 (3:50),
“Anthropic tells investors annualized
revenue run rate climbed to \$65
billion in July”

Example, Is Anthropic Profitable?

- Is Anthropic profitable in Q2 2026?
- We will have a clear understanding after this module



CNBC, August 18, 2026 (3:50),
“Anthropic tells investors annualized
revenue run rate climbed to \$65
billion in July”

Three Financial Statements

- **Balance sheet**
 - What do firms own and owe at a point in time



Three Financial Statements

- **Balance sheet**
 - What do firms own and owe at a point in time
- **Income statement**
 - How much did firms earn in the past year



Three Financial Statements

- **Balance sheet**
 - What do firms own and owe at a point in time
- **Income statement**
 - How much did firms earn in the past year
- **Statement of cash flows**
 - Cash inflows and outflows during the past year



Securities and Exchange Commission (SEC)

- **SEC**, the federal agency that regulates U.S. securities markets and enforces disclosure



sec.gov/edgar/search

Securities and Exchange Commission (SEC)

- **SEC**, the federal agency that regulates U.S. securities markets and enforces disclosure
- **EDGAR**, the SEC's free public database of company filings



sec.gov/edgar/search

Securities and Exchange Commission (SEC)

- **SEC**, the federal agency that regulates U.S. securities markets and enforces disclosure
- **EDGAR**, the SEC's free public database of company filings
- **10-K**, a firm's audited annual report filed with the SEC



sec.gov/edgar/search

Goals

- Interpret the balance sheet, income statement, and statement of cash flows

Goals

- Interpret the balance sheet, income statement, and statement of cash flows
- Distinguish between market and book values

Goals

- Interpret the balance sheet, income statement, and statement of cash flows
- Distinguish between market and book values
- Explain why income differs from cash flow

Goals

- Interpret the balance sheet, income statement, and statement of cash flows
- Distinguish between market and book values
- Explain why income differs from cash flow
- Understand the taxation of corporate and personal income

Module 2, Part 1

The Balance Sheet

Balance Sheet

- **Balance sheet**, assets, liabilities, and equity at a point in time

Balance Sheet

- **Balance sheet**, assets, liabilities, and equity at a point in time
- Components
 - **Assets**, what the firm owns and uses to generate return

Balance Sheet

- **Balance sheet**, assets, liabilities, and equity at a point in time
- Components
 - **Assets**, what the firm owns and uses to generate return
 - **Liabilities**, what the firm owes to others

Balance Sheet

- **Balance sheet**, assets, liabilities, and equity at a point in time
- Components
 - **Assets**, what the firm owns and uses to generate return
 - **Liabilities**, what the firm owes to others
 - **Shareholders' equity**, a residual term that equals assets minus liabilities

Balance Sheet

- **Balance sheet**, assets, liabilities, and equity at a point in time
- Components
 - **Assets**, what the firm owns and uses to generate return
 - **Liabilities**, what the firm owes to others
 - **Shareholders' equity**, a residual term that equals assets minus liabilities

$$\text{Assets} = \text{Liabilities} + \text{Equity}$$

Balance Sheet

Where does the balance sheet come from?

- Recall from **Module 1**, businesses sell financial assets (financing) to purchase real assets (investment)

Balance Sheet

Where does the balance sheet come from?

- Recall from **Module 1**, businesses sell financial assets (financing) to purchase real assets (investment)
- The balance sheet “grows” in this process

Balance Sheet

Where does the balance sheet come from?

- Recall from **Module 1**, businesses sell financial assets (financing) to purchase real assets (investment)
- The balance sheet “grows” in this process
- The numbers are the cost of acquiring real assets, or the amount raised from selling financial assets

Balance Sheet

Where does the balance sheet come from?

- Recall from **Module 1**, businesses sell financial assets (financing) to purchase real assets (investment)
- The balance sheet “grows” in this process
- The numbers are the cost of acquiring real assets, or the amount raised from selling financial assets
- The numbers stay at the transaction amount, even when the asset is worth more or less later

Example, Issuing Stock to Build a Data Center (\$ Billions)

1. Today			
Cash	1	Liabilities	0
Fixed assets	0	Shareholders' equity	1
Total assets	1	Total liabilities and equity	1

Example, Issuing Stock to Build a Data Center (\$ Billions)

1. Today

Cash	1	Liabilities	0
Fixed assets	0	Shareholders' equity	1
Total assets	1	Total liabilities and equity	1

2. Issue stock

Cash	2	Liabilities	0
Fixed assets	0	Shareholders' equity	2
Total assets	2	Total liabilities and equity	2

Example, Issuing Stock to Build a Data Center (\$ Billions)

1. Today			
Cash	1	Liabilities	0
Fixed assets	0	Shareholders' equity	1
Total assets	1	Total liabilities and equity	1

2. Issue stock			
Cash	2	Liabilities	0
Fixed assets	0	Shareholders' equity	2
Total assets	2	Total liabilities and equity	2

3. Build data center			
Cash	0	Liabilities	0
Fixed assets	2	Shareholders' equity	2
Total assets	2	Total liabilities and equity	2

Example, Issuing Stock to Build a Data Center (\$ Billions)

1. Today			
Cash	1	Liabilities	0
Fixed assets	0	Shareholders' equity	1
Total assets	1	Total liabilities and equity	1

2. Issue stock			
Cash	2	Liabilities	0
Fixed assets	0	Shareholders' equity	2
Total assets	2	Total liabilities and equity	2

3. Build data center			
Cash	0	Liabilities	0
Fixed assets	2	Shareholders' equity	2
Total assets	2	Total liabilities and equity	2

- *Should you revise fixed assets if the data center is now worth more than \$2 billion?*

Example, Issuing Stock to Build a Data Center (\$ Billions)

1. Today			
Cash	1	Liabilities	0
Fixed assets	0	Shareholders' equity	1
Total assets	1	Total liabilities and equity	1

2. Issue stock			
Cash	2	Liabilities	0
Fixed assets	0	Shareholders' equity	2
Total assets	2	Total liabilities and equity	2

3. Build data center			
Cash	0	Liabilities	0
Fixed assets	2	Shareholders' equity	2
Total assets	2	Total liabilities and equity	2

- *Should you revise fixed assets if the data center is now worth more than \$2 billion? **No***

Example, Issuing Stock to Build a Data Center (\$ Billions)

1. Today			
Cash	1	Liabilities	0
Fixed assets	0	Shareholders' equity	1
Total assets	1	Total liabilities and equity	1

2. Issue stock			
Cash	2	Liabilities	0
Fixed assets	0	Shareholders' equity	2
Total assets	2	Total liabilities and equity	2

3. Build data center			
Cash	0	Liabilities	0
Fixed assets	2	Shareholders' equity	2
Total assets	2	Total liabilities and equity	2

- *Should you revise fixed assets if the data center is now worth more than \$2 billion? **No***
- *Should you revise shareholders' equity if someone is willing to buy the business at \$4 billion?*

Example, Issuing Stock to Build a Data Center (\$ Billions)

1. Today

Cash	1	Liabilities	0
Fixed assets	0	Shareholders' equity	1
Total assets	1	Total liabilities and equity	1

2. Issue stock

Cash	2	Liabilities	0
Fixed assets	0	Shareholders' equity	2
Total assets	2	Total liabilities and equity	2

3. Build data center

Cash	0	Liabilities	0
Fixed assets	2	Shareholders' equity	2
Total assets	2	Total liabilities and equity	2

- *Should you revise fixed assets if the data center is now worth more than \$2 billion? **No***
- *Should you revise shareholders' equity if someone is willing to buy the business at \$4 billion? **No***

Balance Sheet

Where does the balance sheet come from?

- Recall from **Module 1**, businesses sell financial assets (financing) to purchase real assets (investment)
- The balance sheet “grows” in this process
- The numbers are the cost of acquiring real assets, or the amount raised from selling financial assets
- The numbers stay at the transaction amount, even when the asset is worth more or less later

Breaking Down the Balance Sheet of a Business (\$ Millions)

Current assets

Assets that can generate return within one year

Assets

Current assets

Cash and marketable securities	2,600
Accounts receivable	500
Inventories	9,000
Total current assets	12,100

Fixed assets

Tangible fixed assets

Property, plant, equipment	48,200
Less accumulated depreciation	19,700
Net tangible fixed assets	28,500

Intangible asset	700
------------------	-----

Total fixed assets	29,200
--------------------	--------

Total assets	41,300
---------------------	---------------

Breaking Down the Balance Sheet of a Business (\$ Millions)

Current assets

Assets that can generate return within one year

1. Cash and marketable securities

Assets

Current assets

Cash and marketable securities	2,600
Accounts receivable	500
Inventories	9,000
Total current assets	12,100

Fixed assets

Tangible fixed assets	
Property, plant, equipment	48,200
Less accumulated depreciation	19,700
Net tangible fixed assets	28,500
Intangible asset	700
Total fixed assets	29,200
Total assets	41,300

Breaking Down the Balance Sheet of a Business (\$ Millions)

Current assets

Assets that can generate return within one year

1. Cash and marketable securities
2. Accounts receivable
 - Amount owed by customers

Assets

Current assets

Cash and marketable securities	2,600
--------------------------------	-------

Accounts receivable	500
---------------------	-----

Inventories	9,000
-------------	-------

Total current assets	12,100
----------------------	--------

Fixed assets

Tangible fixed assets

Property, plant, equipment	48,200
----------------------------	--------

Less accumulated depreciation	19,700
-------------------------------	--------

Net tangible fixed assets	28,500
---------------------------	--------

Intangible asset	700
------------------	-----

Total fixed assets	29,200
--------------------	--------

Total assets	41,300
---------------------	---------------

Breaking Down the Balance Sheet of a Business (\$ Millions)

Current assets

Assets that can generate return within one year

1. Cash and marketable securities
2. Accounts receivable
 - Amount owed by customers
3. Inventories
 - The amount in the table is always the acquisition cost

Assets

Current assets

Cash and marketable securities	2,600
Accounts receivable	500

Inventories	9,000
--------------------	--------------

Total current assets	12,100
----------------------	--------

Fixed assets

Tangible fixed assets

Property, plant, equipment	48,200
Less accumulated depreciation	19,700

Net tangible fixed assets	28,500
---------------------------	--------

Intangible asset	700
------------------	-----

Total fixed assets	29,200
--------------------	--------

Total assets	41,300
---------------------	---------------

Breaking Down the Balance Sheet of a Business (\$ Millions)

Current assets

Assets that can generate return within one year

1. Cash and marketable securities
2. Accounts receivable
 - Amount owed by customers
3. Inventories
 - The amount in the table is always the acquisition cost

Assets

Current assets

Cash and marketable securities	2,600
Accounts receivable	500
Inventories	9,000

Total current assets	12,100
-----------------------------	---------------

Fixed assets

Tangible fixed assets

Property, plant, equipment	48,200
Less accumulated depreciation	19,700
Net tangible fixed assets	28,500

Intangible asset	700
------------------	-----

Total fixed assets	29,200
--------------------	--------

Total assets	41,300
---------------------	---------------

Breaking Down the Balance Sheet of a Business (\$ Millions)

Fixed assets

Assets that can generate return for longer than one year

Assets

Current assets

Cash and marketable securities	2,600
--------------------------------	-------

Accounts receivable	500
---------------------	-----

Inventories	9,000
-------------	-------

Total current assets	12,100
----------------------	--------

Fixed assets

Tangible fixed assets

Property, plant, equipment	48,200
----------------------------	--------

Less accumulated depreciation	19,700
-------------------------------	--------

Net tangible fixed assets	28,500
---------------------------	--------

Intangible asset	700
------------------	-----

Total fixed assets	29,200
--------------------	--------

Total assets	41,300
---------------------	---------------

Breaking Down the Balance Sheet of a Business (\$ Millions)

Fixed assets

Assets that can generate return for longer than one year

1. Tangible fixed assets

- Property, plant, equipment ("PP&E"), the amount in the table is always the acquisition cost
- Accumulated depreciation, reduction in the value of assets over time

Assets

Current assets

Cash and marketable securities	2,600
Accounts receivable	500
Inventories	9,000
Total current assets	12,100

Fixed assets

Tangible fixed assets

Property, plant, equipment	48,200
Less accumulated depreciation	19,700

Net tangible fixed assets 28,500

Intangible asset 700

Total fixed assets 29,200

Total assets 41,300

Breaking Down the Balance Sheet of a Business (\$ Millions)

Fixed assets

Assets that can generate return for longer than one year

1. Tangible fixed assets

- Property, plant, equipment ("PP&E"), the amount in the table is always the acquisition cost
- Accumulated depreciation, reduction in the value of assets over time

Assets

Current assets

Cash and marketable securities	2,600
Accounts receivable	500
Inventories	9,000
Total current assets	12,100

Fixed assets

Tangible fixed assets

Property, plant, equipment	48,200
Less accumulated depreciation	19,700

Net tangible fixed assets	28,500
----------------------------------	---------------

Intangible asset	700
------------------	-----

Total fixed assets	29,200
--------------------	--------

Total assets	41,300
---------------------	---------------

Breaking Down the Balance Sheet of a Business (\$ Millions)

Fixed assets

Assets that can generate return for longer than one year

1. Tangible fixed assets
 - Property, plant, equipment (“PP&E”), the amount in the table is always the acquisition cost
 - Accumulated depreciation, reduction in the value of assets over time
2. Intangible assets
 - Difficult to value them accurately

Assets

Current assets

Cash and marketable securities	2,600
Accounts receivable	500
Inventories	9,000
Total current assets	12,100

Fixed assets

Tangible fixed assets

Property, plant, equipment	48,200
Less accumulated depreciation	19,700
Net tangible fixed assets	28,500

Intangible asset	700
------------------	-----

Total fixed assets	29,200
--------------------	--------

Total assets	41,300
---------------------	---------------

Breaking Down the Balance Sheet of a Business (\$ Millions)

Fixed assets

Assets that can generate return for longer than one year

1. Tangible fixed assets
 - Property, plant, equipment (“PP&E”), the amount in the table is always the acquisition cost
 - Accumulated depreciation, reduction in the value of assets over time
2. Intangible assets
 - Difficult to value them accurately

Assets

Current assets

Cash and marketable securities	2,600
Accounts receivable	500
Inventories	9,000
Total current assets	12,100

Fixed assets

Tangible fixed assets

Property, plant, equipment	48,200
Less accumulated depreciation	19,700
Net tangible fixed assets	28,500

Intangible asset	700
------------------	-----

Total fixed assets	29,200
---------------------------	---------------

Total assets	41,300
---------------------	---------------

Breaking Down the Balance Sheet of a Business (\$ Millions)

Fixed assets

Assets that can generate return for longer than one year

1. Tangible fixed assets

- Property, plant, equipment ("PP&E"), the amount in the table is always the acquisition cost
- Accumulated depreciation, reduction in the value of assets over time

2. Intangible assets

- Difficult to value them accurately

Assets

Current assets

Cash and marketable securities	2,600
Accounts receivable	500
Inventories	9,000
Total current assets	12,100

Fixed assets

Tangible fixed assets

Property, plant, equipment	48,200
Less accumulated depreciation	19,700
Net tangible fixed assets	28,500

Intangible asset	700
Total fixed assets	29,200

Total assets	41,300
---------------------	---------------

Breaking Down the Balance Sheet of a Business (\$ Millions)

Current liabilities

Obligations due within a year

Liabilities and equity

Current liabilities

Debt due for repayment	200
------------------------	-----

Accounts payable	9,900
------------------	-------

Total current liabilities	10,100
---------------------------	--------

Long-term liabilities	11,300
-----------------------	--------

Total liabilities	21,400
-------------------	--------

Shareholders' equity

Paid-in capital	6,300
-----------------	-------

Retained earnings	13,600
-------------------	--------

Total shareholders' equity	19,900
----------------------------	--------

Total liabilities and equity	41,300
-------------------------------------	---------------

Breaking Down the Balance Sheet of a Business (\$ Millions)

Current liabilities

Obligations due within a year

1. Debt due for repayment
 - Debt due for repayment within a year

Liabilities and equity

Current liabilities

Debt due for repayment	200
------------------------	-----

Accounts payable	9,900
------------------	-------

Total current liabilities	10,100
---------------------------	--------

Long-term liabilities	11,300
-----------------------	--------

Total liabilities	21,400
-------------------	--------

Shareholders' equity

Paid-in capital	6,300
-----------------	-------

Retained earnings	13,600
-------------------	--------

Total shareholders' equity	19,900
----------------------------	--------

Total liabilities and equity	41,300
-------------------------------------	---------------

Breaking Down the Balance Sheet of a Business (\$ Millions)

Current liabilities

Obligations due within a year

1. Debt due for repayment
 - Debt due for repayment within a year
2. Accounts payable
 - Amount owed to suppliers

Liabilities and equity

Current liabilities

Debt due for repayment	200
------------------------	-----

Accounts payable	9,900
------------------	-------

Total current liabilities	10,100
---------------------------	--------

Long-term liabilities	11,300
-----------------------	--------

Total liabilities	21,400
-------------------	--------

Shareholders' equity

Paid-in capital	6,300
-----------------	-------

Retained earnings	13,600
-------------------	--------

Total shareholders' equity	19,900
----------------------------	--------

Total liabilities and equity	41,300
-------------------------------------	---------------

Breaking Down the Balance Sheet of a Business (\$ Millions)

Current liabilities

Obligations due within a year

1. Debt due for repayment
 - Debt due for repayment within a year
2. Accounts payable
 - Amount owed to suppliers

Liabilities and equity

Current liabilities

Debt due for repayment	200
Accounts payable	9,900

Total current liabilities	10,100
----------------------------------	---------------

Long-term liabilities	11,300
-----------------------	--------

Total liabilities	21,400
-------------------	--------

Shareholders' equity

Paid-in capital	6,300
Retained earnings	13,600

Total shareholders' equity	19,900
----------------------------	--------

Total liabilities and equity	41,300
-------------------------------------	---------------

Breaking Down the Balance Sheet of a Business (\$ Millions)

Long-term liabilities

Obligations due after a year

Liabilities and equity

Current liabilities

Debt due for repayment	200
------------------------	-----

Accounts payable	9,900
------------------	-------

Total current liabilities	10,100
---------------------------	--------

Long-term liabilities	11,300
-----------------------	--------

Total liabilities	21,400
-------------------	--------

Shareholders' equity

Paid-in capital	6,300
-----------------	-------

Retained earnings	13,600
-------------------	--------

Total shareholders' equity	19,900
----------------------------	--------

Total liabilities and equity	41,300
-------------------------------------	---------------

Breaking Down the Balance Sheet of a Business (\$ Millions)

Long-term liabilities

Obligations due after a year

1. Long-term debt

- Bonds, debt from public bond markets
- Bank loans, debt from banks

Liabilities and equity

Current liabilities

Debt due for repayment	200
------------------------	-----

Accounts payable	9,900
------------------	-------

Total current liabilities	10,100
---------------------------	--------

Long-term liabilities	11,300
-----------------------	--------

Total liabilities	21,400
-------------------	--------

Shareholders' equity

Paid-in capital	6,300
-----------------	-------

Retained earnings	13,600
-------------------	--------

Total shareholders' equity	19,900
----------------------------	--------

Total liabilities and equity	41,300
-------------------------------------	---------------

Breaking Down the Balance Sheet of a Business (\$ Millions)

Long-term liabilities

Obligations due after a year

1. Long-term debt

- Bonds, debt from public bond markets
- Bank loans, debt from banks

Liabilities and equity

Current liabilities

Debt due for repayment	200
------------------------	-----

Accounts payable	9,900
------------------	-------

Total current liabilities	10,100
---------------------------	--------

Long-term liabilities	11,300
-----------------------	--------

Total liabilities	21,400
--------------------------	---------------

Shareholders' equity

Paid-in capital	6,300
-----------------	-------

Retained earnings	13,600
-------------------	--------

Total shareholders' equity	19,900
----------------------------	--------

Total liabilities and equity	41,300
-------------------------------------	---------------

Breaking Down the Balance Sheet of a Business (\$ Millions)

Shareholders' equity

A *residual* term that equals assets minus liabilities

Liabilities and equity

Current liabilities	
Debt due for repayment	200
Accounts payable	9,900
Total current liabilities	<u>10,100</u>
Long-term liabilities	11,300
Total liabilities	<u>21,400</u>
Shareholders' equity	
Paid-in capital	6,300
Retained earnings	13,600
Total shareholders' equity	<u>19,900</u>
Total liabilities and equity	<u>41,300</u>

Breaking Down the Balance Sheet of a Business (\$ Millions)

Shareholders' equity

A *residual* term that equals assets minus liabilities

1. Paid-in capital
 - Amount raised from previous sales of ownership

Liabilities and equity

Current liabilities	
Debt due for repayment	200
Accounts payable	9,900
Total current liabilities	10,100
Long-term liabilities	11,300
Total liabilities	21,400

Shareholders' equity

Paid-in capital	6,300
Retained earnings	13,600
Total shareholders' equity	19,900

Total liabilities and equity	41,300
-------------------------------------	---------------

Breaking Down the Balance Sheet of a Business (\$ Millions)

Shareholders' equity

A *residual* term that equals assets minus liabilities

1. Paid-in capital
 - Amount raised from previous sales of ownership
2. Retained earnings
 - Accumulated profits kept in the business and reinvested on the shareholders' behalf

Liabilities and equity

Current liabilities	
Debt due for repayment	200
Accounts payable	9,900
Total current liabilities	10,100
Long-term liabilities	11,300
Total liabilities	21,400
Shareholders' equity	
Paid-in capital	6,300
Retained earnings	13,600
Total shareholders' equity	19,900
Total liabilities and equity	41,300

Breaking Down the Balance Sheet of a Business (\$ Millions)

Shareholders' equity

A *residual* term that equals assets minus liabilities

1. Paid-in capital
 - Amount raised from previous sales of ownership
2. Retained earnings
 - Accumulated profits kept in the business and reinvested on the shareholders' behalf

Liabilities and equity

Current liabilities	
Debt due for repayment	200
Accounts payable	9,900
Total current liabilities	10,100
Long-term liabilities	11,300
Total liabilities	21,400

Shareholders' equity	
Paid-in capital	6,300
Retained earnings	13,600
Total shareholders' equity	19,900

Total liabilities and equity	41,300
-------------------------------------	---------------

Breaking Down the Balance Sheet of a Business (\$ Millions)

Shareholders' equity

A *residual* term that equals assets minus liabilities

1. Paid-in capital
 - Amount raised from previous sales of ownership
2. Retained earnings
 - Accumulated profits kept in the business and reinvested on the shareholders' behalf

Liabilities and equity

Current liabilities	
Debt due for repayment	200
Accounts payable	9,900
Total current liabilities	<u>10,100</u>
Long-term liabilities	11,300
Total liabilities	<u>21,400</u>

Shareholders' equity	
Paid-in capital	6,300
Retained earnings	13,600
Total shareholders' equity	<u>19,900</u>

Total liabilities and equity	<u>41,300</u>
-------------------------------------	----------------------

Breaking Down the Balance Sheet of a Business (\$ Millions)

Most liquid 	Assets		Liabilities and equity		Paid first 
	Current assets		Current liabilities		
	Cash and marketable sec.	2,600	Debt due for repayment	200	
	Accounts receivable	500	Accounts payable	9,900	
	Inventories	9,000	Total current liabilities	10,100	
	Total current assets	12,100	Long-term liabilities	11,300	
	Fixed assets		Total liabilities		
	Tangible fixed assets				
	Property, plant, equipment	48,200	Shareholders' equity		
	Less accum. depreciation	19,700	Paid-in capital	6,300	
Least liquid	Net tangible fixed assets	28,500	Retained earnings	13,600	Paid last
	Intangible asset	700	Total shareholders' equity	19,900	
	Total fixed assets	29,200			
	Total assets	41,300	Total liabilities and equity	41,300	

- **Liquidity**, the ease of converting an asset into cash

Book Value vs. Market Value

1. **Book values**, the cost of acquiring real assets, or the amount raised from selling financial assets
 - Values on balance sheets are book value

Book Value vs. Market Value

1. **Book values**, the cost of acquiring real assets, or the amount raised from selling financial assets
 - Values on balance sheets are book value
 - E.g., the \$2 billion data center built last year → book value, forever

Book Value vs. Market Value

1. **Book values**, the cost of acquiring real assets, or the amount raised from selling financial assets
 - Values on balance sheets are book value
 - E.g., the \$2 billion data center built last year → book value, forever
2. **Market values**, what things would sell for today
 - E.g., worth \$4 billion today → market value

Book Value vs. Market Value

1. **Book values**, the cost of acquiring real assets, or the amount raised from selling financial assets
 - Values on balance sheets are book value
 - E.g., the \$2 billion data center built last year → book value, forever
 2. **Market values**, what things would sell for today
 - E.g., worth \$4 billion today → market value
- In general, book $\neq$ market value

Example, Issuing Stock to Build a Data Center (\$ Billions)

1. Today			
Cash	1	Liabilities	0
Fixed assets	0	Shareholders' equity	1
Total assets	1	Total liabilities and equity	1

2. Issue stock			
Cash	2	Liabilities	0
Fixed assets	0	Shareholders' equity	2
Total assets	2	Total liabilities and equity	2

3. Build data center			
Cash	0	Liabilities	0
Fixed assets	2	Shareholders' equity	2
Total assets	2	Total liabilities and equity	2

- *Should you revise fixed assets if the data center is now worth more than \$2 billion? **No***
- *Should you revise shareholders' equity if someone is willing to buy the business at \$4 billion? **No***

Book Value vs. Market Value

1. Book value of equity = Book value of assets – Book value of liabilities
 - E.g., Apple at the end of fiscal year 2025,
 - Book value of assets, \$359,241 Mn
 - Book value of liabilities, \$285,508 Mn
 - What is the book value of equity?

Book Value vs. Market Value

1. Book value of equity = Book value of assets – Book value of liabilities
 - E.g., Apple at the end of fiscal year 2025,
 - Book value of assets, \$359,241 Mn
 - Book value of liabilities, \$285,508 Mn
 - What is the book value of equity?
 - Answer, $\$359,241 \text{ Mn} - \$285,508 \text{ Mn} = \$73,733 \text{ Mn}$

Book Value vs. Market Value

1. Book value of equity = Book value of assets – Book value of liabilities
 - E.g., Apple at the end of fiscal year 2025,
 - Book value of assets, \$359,241 Mn
 - Book value of liabilities, \$285,508 Mn
 - What is the book value of equity?
 - Answer, $\$359,241 \text{ Mn} - \$285,508 \text{ Mn} = \$73,733 \text{ Mn}$
2. Market value of equity = Market price of each share $\times$ Total shares outstanding
 - E.g., Apple at the end of 2025,
 - Share price, about \$271
 - Shares outstanding, 14,800 Mn
 - What is the market value of equity?

Book Value vs. Market Value

1. Book value of equity = Book value of assets – Book value of liabilities
 - E.g., Apple at the end of fiscal year 2025,
 - Book value of assets, \$359,241 Mn
 - Book value of liabilities, \$285,508 Mn
 - What is the book value of equity?
 - Answer, $\$359,241 \text{ Mn} - \$285,508 \text{ Mn} = \$73,733 \text{ Mn}$
2. Market value of equity = Market price of each share $\times$ Total shares outstanding
 - E.g., Apple at the end of 2025,
 - Share price, about \$271
 - Shares outstanding, 14,800 Mn
 - What is the market value of equity?
 - Answer, $\$271 \times 14,800 \text{ Mn} = \$4,010,800 \text{ Mn}$

Book Value vs. Market Value

- Book value of equity reflects the money that equity investors have invested into the firm



Book Value vs. Market Value

- Book value of equity reflects the money that equity investors have invested into the firm
- Market value of equity reflects the money investors expect the firm to make in the future



Book Value vs. Market Value

- Book value of equity reflects the money that equity investors have invested into the firm
- Market value of equity reflects the money investors expect the firm to make in the future
- Naturally, market value of equity $>$ book value of equity for a successful business



Practice Problem

- A business's book value of total assets is \$45,000. Its book value of current liabilities is \$15,000, and its book value of shareholders' equity is \$20,000. How much is its book value of long-term liabilities?

Practice Problem

- A business's book value of total assets is \$45,000. Its book value of current liabilities is \$15,000, and its book value of shareholders' equity is \$20,000. How much is its book value of long-term liabilities?
- $\text{Assets} = \text{Liabilities} + \text{Equity}$

Practice Problem

- A business's book value of total assets is \$45,000. Its book value of current liabilities is \$15,000, and its book value of shareholders' equity is \$20,000. How much is its book value of long-term liabilities?
- $\text{Assets} = \text{Liabilities} + \text{Equity}$
- $\$45,000 = \$15,000 + \text{_____} + \$20,000$

Practice Problem

- A business's book value of total assets is \$45,000. Its book value of current liabilities is \$15,000, and its book value of shareholders' equity is \$20,000. How much is its book value of long-term liabilities?
- $\text{Assets} = \text{Liabilities} + \text{Equity}$
- $\$45,000 = \$15,000 + \text{_____} + \$20,000$
- $\$45,000 = \$15,000 + \$10,000 + \$20,000$

Practice Problem

- Which of the following are current assets? Select all that apply.
 - a. Cash.
 - b. A software license (3-year term).
 - c. Wages payable (due in 2 weeks).
 - d. A bank loan (due in 3 years).
 - e. Accounts receivable.
 - f. An office building.

Practice Problem

- Which of the following are current assets? Select all that apply.
 - a. Cash.
 - b. A software license (3-year term).
 - c. Wages payable (due in 2 weeks).
 - d. A bank loan (due in 3 years).
 - e. Accounts receivable.
 - f. An office building.

Recap of Part 1

- Accounting conveys information to investors and stakeholders
- The balance sheet, assets, liabilities, and equity at a point in time
- $\text{Assets} = \text{Liabilities} + \text{Equity}$
- In general, book $\neq$ market value

Module 2, Part 2

The Income Statement

Income Statement

- **Income statement**, how much firms earned in the past year



INCOME STATEMENTS

Income Statement

- **Revenues**, sales made in the period, whether or not cash was collected
 - E.g., a firm sells 100 products to customers, each at a price of \$100
 - $\text{Revenues} = 100 \times \$100 = \$10,000$

	\$ Million
Revenues	80,000
COGS	56,000
SG&A expenses	16,000
Depreciation	3,000
EBIT	5,000
Interest expense	1,000
Taxable income	4,000
Taxes	840
Net income	3,160
Allocation of net income	
Dividends	1,000
Addition to retained earnings	2,160

Income Statement

- **Operating expenses**, the costs of running the business

	\$ Million
Revenues	80,000
COGS	56,000
SG&A expenses	16,000
Depreciation	3,000
EBIT	5,000
Interest expense	1,000
Taxable income	4,000
Taxes	840
Net income	3,160
Allocation of net income	
Dividends	1,000
Addition to retained earnings	2,160

Income Statement

- **Operating expenses**, the costs of running the business
 - **Cost of goods sold (COGS)**, the direct cost of the goods actually sold
 - E.g., each of the 100 products sold costs the firm \$60 to make
 - $\text{COGS} = 100 \times \$60 = \$6,000$

	\$ Million
Revenues	80,000
COGS	56,000
SG&A expenses	16,000
Depreciation	3,000
EBIT	5,000
Interest expense	1,000
Taxable income	4,000
Taxes	840
Net income	3,160
Allocation of net income	
Dividends	1,000
Addition to retained earnings	2,160

Income Statement

- **Operating expenses**, the costs of running the business
 - **Cost of goods sold (COGS)**, the direct cost of the goods actually sold
 - E.g., each of the 100 products sold costs the firm \$60 to make
 - $\text{COGS} = 100 \times \$60 = \$6,000$
 - **Selling, general & administrative (SG&A)**, other operating costs, e.g., marketing, salaries, distribution

	\$ Million
Revenues	80,000
COGS	56,000
SG&A expenses	16,000
Depreciation	3,000
EBIT	5,000
Interest expense	1,000
Taxable income	4,000
Taxes	840
Net income	3,160
Allocation of net income	
Dividends	1,000
Addition to retained earnings	2,160

Income Statement

- **Operating expenses**, the costs of running the business
 - **Cost of goods sold (COGS)**, the direct cost of the goods actually sold
 - E.g., each of the 100 products sold costs the firm \$60 to make
 - $COGS = 100 \times \$60 = \$6,000$
 - **Selling, general & administrative (SG&A)**, other operating costs, e.g., marketing, salaries, distribution
 - **Depreciation**, this period's share of a fixed asset's cost

	\$ Million
Revenues	80,000
COGS	56,000
SG&A expenses	16,000
Depreciation	3,000
EBIT	5,000
Interest expense	1,000
Taxable income	4,000
Taxes	840
Net income	3,160
Allocation of net income	
Dividends	1,000
Addition to retained earnings	2,160

Fixed Assets Spread over Time as Depreciation

- Fixed assets earn revenue for years, so we spread their cost over those years instead of subtracting it all at once



Fixed Assets Spread over Time as Depreciation

- Fixed assets earn revenue for years, so we spread their cost over those years instead of subtracting it all at once
- This spreading is **depreciation**



Fixed Assets Spread over Time as Depreciation

- Fixed assets earn revenue for years, so we spread their cost over those years instead of subtracting it all at once
- This spreading is **depreciation**
- Two things happen with depreciation,
 - In the income statement, subtract depreciation from revenues as a non-cash expense



Fixed Assets Spread over Time as Depreciation

- Fixed assets earn revenue for years, so we spread their cost over those years instead of subtracting it all at once
- This spreading is **depreciation**
- Two things happen with depreciation,
 - In the income statement, subtract depreciation from revenues as a non-cash expense
 - In the **balance sheet**, add to accumulated depreciation



Fixed Assets Spread over Time as Depreciation

- The firm pays \$2 billion to build a data center that lasts 5 years

Fixed Assets Spread over Time as Depreciation

- The firm pays \$2 billion to build a data center that lasts 5 years
- We spread the cost over these 5 years

Fixed Assets Spread over Time as Depreciation

- The firm pays \$2 billion to build a data center that lasts 5 years
- We spread the cost over these 5 years
- Each year, the depreciation is \$0.4 billion

Fixed Assets Spread over Time as Depreciation

- The firm pays \$2 billion to build a data center that lasts 5 years
- We spread the cost over these 5 years
- Each year, the depreciation is \$0.4 billion

Year	0	1	2	3	4	5	
Cash flow (\$ billions)	-2.0	0	0	0	0	0	
Effect on net income	0	-0.4	-0.4	-0.4	-0.4	-0.4	← Depreciation

Income Statement

- **EBIT**, earnings before interest and taxes
 - $\text{EBIT} = \text{Revenues} - \text{COGS} - \text{SG\&A expenses} - \text{Depreciation}$

	\$ Million
Revenues	80,000
COGS	56,000
SG&A expenses	16,000
Depreciation	3,000
EBIT	5,000
Interest expense	1,000
Taxable income	4,000
Taxes	840
Net income	3,160
Allocation of net income	
Dividends	1,000
Addition to retained earnings	2,160

Income Statement

- **Interest expense**, the interest paid on the firm's debt for the period

	\$ Million
Revenues	80,000
COGS	56,000
SG&A expenses	16,000
Depreciation	3,000
EBIT	5,000
Interest expense	1,000
Taxable income	4,000
Taxes	840
Net income	3,160
Allocation of net income	
Dividends	1,000
Addition to retained earnings	2,160

Income Statement

- **Taxes**

- Taxable income = EBIT – Interest expense
- Taxes = Taxable income × Tax rate

	\$ Million
Revenues	80,000
COGS	56,000
SG&A expenses	16,000
Depreciation	3,000
EBIT	5,000
Interest expense	1,000
Taxable income	4,000
Taxes	840
Net income	3,160
Allocation of net income	
Dividends	1,000
Addition to retained earnings	2,160

Income Statement

- **Net income**, what the business earned in the period after every expense

	\$ Million
Revenues	80,000
COGS	56,000
SG&A expenses	16,000
Depreciation	3,000
EBIT	5,000
Interest expense	1,000
Taxable income	4,000
Taxes	840
Net income	3,160
Allocation of net income	
Dividends	1,000
Addition to retained earnings	2,160

Where Net Income Goes

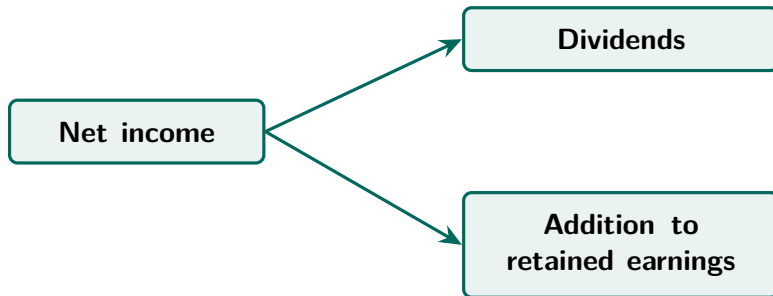
- Shareholders have claims to net income

Where Net Income Goes

- Shareholders have claims to net income
- Net income can be kept in the business, or paid out to shareholders

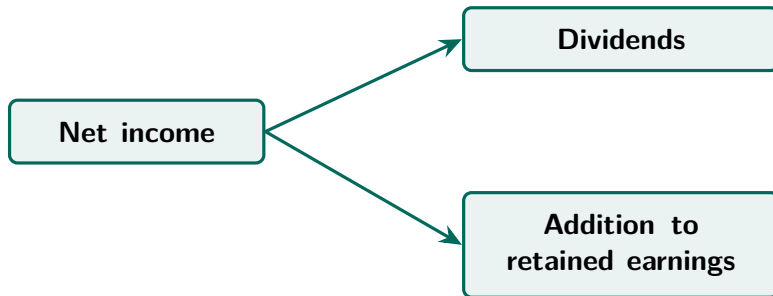
Where Net Income Goes

- Shareholders have claims to net income
- Net income can be kept in the business, or paid out to shareholders



Where Net Income Goes

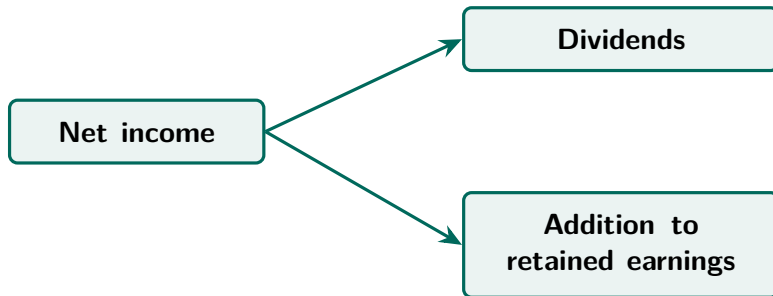
- Shareholders have claims to net income
- Net income can be kept in the business, or paid out to shareholders



- **Dividends**, cash paid out to shareholders now

Where Net Income Goes

- Shareholders have claims to net income
- Net income can be kept in the business, or paid out to shareholders



- **Dividends**, cash paid out to shareholders now
- **Retained earnings**, accumulated profits kept in the business and reinvested on the shareholders' behalf

Income Statement, A Memory Aid

Line item	Letters	Remember it as
Revenues	R	R ainbow
COGS	C	C andy
SG&A expenses	S	S ells
Depreciation	D	D ependably
EBIT	E	E specially
Interest expense	I	I n
Taxable income	T.I.	T exas I ncorporated
Taxes	T	T exas
Net income	N.I.	N ever I ncorporates
Dividends	D	D uring
Addition to retained earnings	R.E.	R e- E lections

Example, Is Anthropic Profitable?

- Is Anthropic profitable in Q2 2026?



CNBC, August 18, 2026 (3:50),
“Anthropic tells investors annualized
revenue run rate climbed to \$65
billion in July”

Example, Is Anthropic Profitable?

- Is Anthropic profitable in Q2 2026?
- Profitable on an **EBITDA** basis
 - EBITDA, earnings before interest, taxes, depreciation, and amortization
 - Amortization, the same cost-spreading as depreciation, applied to intangible assets



CNBC, August 18, 2026 (3:50),
“Anthropic tells investors annualized
revenue run rate climbed to \$65
billion in July”

Example, Is Anthropic Profitable?

- Is Anthropic profitable in Q2 2026?
- Profitable on an **EBITDA** basis
 - EBITDA, earnings before interest, taxes, depreciation, and amortization
 - Amortization, the same cost-spreading as depreciation, applied to intangible assets
- Not profitable on a **net income** basis



CNBC, August 18, 2026 (3:50),
“Anthropic tells investors annualized
revenue run rate climbed to \$65
billion in July”

Practice Problem

	\$
Revenues	?
COGS	20,000
SG&A expenses	10,000
Depreciation	8,000
EBIT	12,000
Interest expense	?
Taxable income	?
Taxes (tax rate = 21%)	2,100
Net income	7,900
Dividends	2,500
Addition to retained earnings	?

Practice Problem

	\$
Revenues	50,000
COGS	20,000
SG&A expenses	10,000
Depreciation	8,000
EBIT	12,000
Interest expense	2,000
Taxable income	10,000
Taxes (tax rate = 21%)	2,100
Net income	7,900
Dividends	2,500
Addition to retained earnings	5,400

Recap of Part 2

- The income statement, how much firms earned in the past year
- **Depreciation**, this period's share of a fixed asset's cost
- Net income can be kept in the business, or paid out to shareholders
- Taxes are computed **after** depreciation and interest, never before

Module 2, Part 3

The Statement of Cash Flows

Statement of Cash Flows

- **Cash flow**, cash actually received minus cash actually paid, no matter when the sale was made

Statement of Cash Flows

- **Cash flow**, cash actually received minus cash actually paid, no matter when the sale was made
- The **statement of cash flows**, cash receipts and payments over a period

Statement of Cash Flows

- **Cash flow**, cash actually received minus cash actually paid, no matter when the sale was made
- The **statement of cash flows**, cash receipts and payments over a period
- The statement has three parts
 - **Operating activities**, cash from running the business day to day

Statement of Cash Flows

- **Cash flow**, cash actually received minus cash actually paid, no matter when the sale was made
- The **statement of cash flows**, cash receipts and payments over a period
- The statement has three parts
 - **Operating activities**, cash from running the business day to day
 - **Investing activities**, cash spent buying fixed assets and cash received from selling them

Statement of Cash Flows

- **Cash flow**, cash actually received minus cash actually paid, no matter when the sale was made
- The **statement of cash flows**, cash receipts and payments over a period
- The statement has three parts
 - **Operating activities**, cash from running the business day to day
 - **Investing activities**, cash spent buying fixed assets and cash received from selling them
 - **Financing activities**, cash raised from lenders and shareholders, and cash returned to them

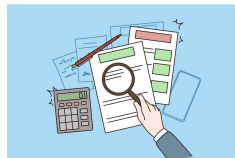
Statement of Cash Flows

Direct method	Indirect method
Lists every cash receipt and payment	Starts from net income and adjusts it
More time-consuming to prepare	Faster to prepare
Rarely used in practice	The common choice

Direct →



Indirect →



Statement of Cash Flows

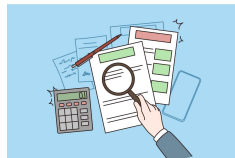
Direct method	Indirect method
Lists every cash receipt and payment	Starts from net income and adjusts it
More time-consuming to prepare	Faster to prepare
Rarely used in practice	The common choice

- Both methods give the same totals

Direct →



Indirect →



Statement of Cash Flows

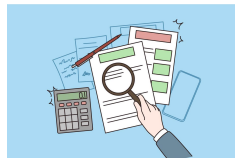
Direct method	Indirect method
Lists every cash receipt and payment	Starts from net income and adjusts it
More time-consuming to prepare	Faster to prepare
Rarely used in practice	The common choice

- Both methods give the same totals
- We will follow the indirect method, starting from net income in the income statement

Direct →



Indirect →



How Do Net Income and Cash Flow Differ?

- The firm pays \$2 billion to build a data center that lasts 5 years

How Do Net Income and Cash Flow Differ?

- The firm pays \$2 billion to build a data center that lasts 5 years
- We spread the cost over these 5 years

How Do Net Income and Cash Flow Differ?

- The firm pays \$2 billion to build a data center that lasts 5 years
- We spread the cost over these 5 years
- Each year, the depreciation is \$0.4 billion

How Do Net Income and Cash Flow Differ?

- The firm pays \$2 billion to build a data center that lasts 5 years
- We spread the cost over these 5 years
- Each year, the depreciation is \$0.4 billion

Year	0	1	2	3	4	5
Cash flow (\$ billions)	-2.0	0	0	0	0	0
Effect on net income	0	-0.4	-0.4	-0.4	-0.4	-0.4

← Depreciation

How Do Net Income and Cash Flow Differ?

- The firm pays \$2 billion to build a data center that lasts 5 years
- We spread the cost over these 5 years
- Each year, the depreciation is \$0.4 billion

Year	0	1	2	3	4	5
Cash flow (\$ billions)	-2.0	0	0	0	0	0
Effect on net income	0	-0.4	-0.4	-0.4	-0.4	-0.4

← Depreciation

- Cash leaves all at once, net income feels it slowly

How Do Net Income and Cash Flow Differ?

- The firm pays \$2 billion to build a data center that lasts 5 years
- We spread the cost over these 5 years
- Each year, the depreciation is \$0.4 billion

Year	0	1	2	3	4	5
Cash flow (\$ billions)	−2.0	0	0	0	0	0
Effect on net income	0	−0.4	−0.4	−0.4	−0.4	−0.4

← Depreciation

- Cash leaves all at once, net income feels it slowly

Relationship of cash flow (CF) and net income (NI)?

How Do Net Income and Cash Flow Differ?

- The firm pays \$2 billion to build a data center that lasts 5 years
- We spread the cost over these 5 years
- Each year, the depreciation is \$0.4 billion

Year	0	1	2	3	4	5
Cash flow (\$ billions)	-2.0	0	0	0	0	0
Effect on net income	0	-0.4	-0.4	-0.4	-0.4	-0.4

← Depreciation

- Cash leaves all at once, net income feels it slowly

Relationship of cash flow (CF) and net income (NI)?

When spending on fixed assets, $CF = NI - \text{spending on fixed assets}$

How Do Net Income and Cash Flow Differ?

- The firm pays \$2 billion to build a data center that lasts 5 years
- We spread the cost over these 5 years
- Each year, the depreciation is \$0.4 billion

Year	0	1	2	3	4	5
Cash flow (\$ billions)	-2.0	0	0	0	0	0
Effect on net income	0	-0.4	-0.4	-0.4	-0.4	-0.4

← Depreciation

- Cash leaves all at once, net income feels it slowly

Relationship of cash flow (CF) and net income (NI)?

When spending on fixed assets, $CF = NI - \text{spending on fixed assets}$

When recording depreciation, $CF = NI + \text{depreciation}$

How Do Net Income and Cash Flow Differ?

- The firm issues (sells) \$1 billion of new stock to investors

How Do Net Income and Cash Flow Differ?

- The firm issues (sells) \$1 billion of new stock to investors

	Effect
Cash flow (\$ billions)	+1.0
Effect on net income	0

How Do Net Income and Cash Flow Differ?

- The firm issues (sells) \$1 billion of new stock to investors

	Effect
Cash flow (\$ billions)	+1.0
Effect on net income	0

- Selling stock is not generating income

How Do Net Income and Cash Flow Differ?

- The firm issues (sells) \$1 billion of new stock to investors

	Effect
Cash flow (\$ billions)	+1.0
Effect on net income	0

- Selling stock is not generating income
- Net income never records money raised from issuing (selling) stocks or bonds or borrowing from banks

How Do Net Income and Cash Flow Differ?

- The firm issues (sells) \$1 billion of new stock to investors

	Effect
Cash flow (\$ billions)	+1.0
Effect on net income	0

- Selling stock is not generating income
- Net income never records money raised from issuing (selling) stocks or bonds or borrowing from banks

When raising money, $CF = NI + \text{amount raised}$

How Do Net Income and Cash Flow Differ?

- The firm spends \$60 million producing AI control panels
- None of them have been sold yet



OpenAI, July 15, 2026 (2:15),
“Introducing the Codex Micro”

How Do Net Income and Cash Flow Differ?

- The firm spends \$60 million producing AI control panels
- None of them have been sold yet

	Effect
Cash flow (\$ millions)	−60
Effect on net income	0
<u>Change in inventories</u>	<u>+60</u>



OpenAI, July 15, 2026 (2:15),
“Introducing the Codex Micro”

How Do Net Income and Cash Flow Differ?

- The firm spends \$60 million producing AI control panels
- None of them have been sold yet

	Effect
Cash flow (\$ millions)	−60
Effect on net income	0
<u>Change in inventories</u>	<u>+60</u>

- The income statement records revenues and cost of goods sold (COGS) at the time of sale



OpenAI, July 15, 2026 (2:15),
“Introducing the Codex Micro”

How Do Net Income and Cash Flow Differ?

- The firm spends \$60 million producing AI control panels
- None of them have been sold yet

	Effect
Cash flow (\$ millions)	−60
Effect on net income	0
<u>Change in inventories</u>	<u>+60</u>

- The income statement records revenues and cost of goods sold (COGS) at the time of sale
- COGS is subtracted only when the panels are sold, not when they are produced



OpenAI, July 15, 2026 (2:15),
“Introducing the Codex Micro”

How Do Net Income and Cash Flow Differ?

- The firm spends \$60 million producing AI control panels
- None of them have been sold yet

	Effect
Cash flow (\$ millions)	−60
Effect on net income	0
<u>Change in inventories</u>	<u>+60</u>

- The income statement records revenues and cost of goods sold (COGS) at the time of sale
- COGS is subtracted only when the panels are sold, not when they are produced

When adding to inventories, $CF = NI - \text{change in inventories}$



OpenAI, July 15, 2026 (2:15),
“Introducing the Codex Micro”

How Do Net Income and Cash Flow Differ?

- The firm sells AI control panels for \$100 million
- The customer has not paid yet



OpenAI, July 15, 2026 (2:15),
“Introducing the Codex Micro”

How Do Net Income and Cash Flow Differ?

- The firm sells AI control panels for \$100 million
- The customer has not paid yet

	Effect
Cash flow (\$ millions)	0
Effect on net income	+100
Change in accounts receivable	+100



OpenAI, July 15, 2026 (2:15),
“Introducing the Codex Micro”

How Do Net Income and Cash Flow Differ?

- The firm sells AI control panels for \$100 million
- The customer has not paid yet

	Effect
Cash flow (\$ millions)	0
Effect on net income	+100
Change in accounts receivable	+100

- Net income rises now, cash flow comes later when customers actually pay



OpenAI, July 15, 2026 (2:15),
“Introducing the Codex Micro”

How Do Net Income and Cash Flow Differ?

- The firm sells AI control panels for \$100 million
- The customer has not paid yet

	Effect
Cash flow (\$ millions)	0
Effect on net income	+100
Change in accounts receivable	+100

- Net income rises now, cash flow comes later when customers actually pay



OpenAI, July 15, 2026 (2:15),
“Introducing the Codex Micro”

When firm not yet paid by customers, $CF = NI - \text{change in accounts receivable}$

How Do Net Income and Cash Flow Differ?

- The firm owed suppliers \$30 million for electronic parts
- The firm pays down the \$30 million



OpenAI, July 15, 2026 (2:15),
“Introducing the Codex Micro”

How Do Net Income and Cash Flow Differ?

- The firm owed suppliers \$30 million for electronic parts
- The firm pays down the \$30 million

	Effect
Cash flow (\$ millions)	−30
Effect on net income	0
<u>Change in accounts payable</u>	<u>−30</u>



OpenAI, July 15, 2026 (2:15),
“Introducing the Codex Micro”

How Do Net Income and Cash Flow Differ?

- The firm owed suppliers \$30 million for electronic parts
- The firm pays down the \$30 million

	Effect
Cash flow (\$ millions)	−30
Effect on net income	0
<u>Change in accounts payable</u>	<u>−30</u>

- No sale of panels happened. No revenues or COGS should be recorded. Net income is 0



OpenAI, July 15, 2026 (2:15),
“Introducing the Codex Micro”

How Do Net Income and Cash Flow Differ?

- The firm owed suppliers \$30 million for electronic parts
- The firm pays down the \$30 million

	Effect
Cash flow (\$ millions)	−30
Effect on net income	0
<u>Change in accounts payable</u>	<u>−30</u>

- No sale of panels happened. No revenues or COGS should be recorded. Net income is 0



OpenAI, July 15, 2026 (2:15),
“Introducing the Codex Micro”

How Do Net Income and Cash Flow Differ?

- The firm owed suppliers \$30 million for electronic parts
- The firm pays down the \$30 million

	Effect
Cash flow (\$ millions)	−30
Effect on net income	0
<u>Change in accounts payable</u>	<u>−30</u>

- No sale of panels happened. No revenues or COGS should be recorded. Net income is 0

When firm pays off suppliers, $CF = NI + \text{change in accounts payable}$



OpenAI, July 15, 2026 (2:15),
“Introducing the Codex Micro”

Operating Activities

Operating activities, cash from running the business day to day

Steps to calculate cash flow from operating activities,

1. **Start** with net income
2. **Add** depreciation
3. **Subtract** change in net working capital (NWC)
 - $\text{NWC} = \text{inventories} + \text{accounts receivable} - \text{accounts payable}$
 - NWC, cash tied up in day-to-day operations, in unsold goods and unpaid bills

Why Subtract Change in Net Working Capital (NWC)?

When adding to inventories, $CF = NI - \text{change in inventories}$

When firm not yet paid by customers, $CF = NI - \text{change in accounts receivable}$

When firm pays off suppliers, $CF = NI + \text{change in accounts payable}$

Why Subtract Change in Net Working Capital (NWC)?

When adding to inventories, $CF = NI - \text{change in inventories}$

When firm not yet paid by customers, $CF = NI - \text{change in accounts receivable}$

When firm pays off suppliers, $CF = NI + \text{change in accounts payable}$

Ignoring other adjustments,

Why Subtract Change in Net Working Capital (NWC)?

When adding to inventories, $CF = NI - \text{change in inventories}$

When firm not yet paid by customers, $CF = NI - \text{change in accounts receivable}$

When firm pays off suppliers, $CF = NI + \text{change in accounts payable}$

Ignoring other adjustments,

$$CF = NI$$

- change in inventories*
- change in accounts receivable*
- + change in accounts payable*

Why Subtract Change in Net Working Capital (NWC)?

When adding to inventories, $CF = NI - \text{change in inventories}$

When firm not yet paid by customers, $CF = NI - \text{change in accounts receivable}$

When firm pays off suppliers, $CF = NI + \text{change in accounts payable}$

Ignoring other adjustments,

$$CF = NI$$

– change in inventories

– change in accounts receivable

+ change in accounts payable

$$CF = NI - \text{change in (inventories + accounts receivable – accounts payable)}$$

Why Subtract Change in Net Working Capital (NWC)?

When adding to inventories, $CF = NI - \text{change in inventories}$

When firm not yet paid by customers, $CF = NI - \text{change in accounts receivable}$

When firm pays off suppliers, $CF = NI + \text{change in accounts payable}$

Ignoring other adjustments,

$$CF = NI$$

– change in inventories

– change in accounts receivable

+ change in accounts payable

$$CF = NI - \text{change in (inventories + accounts receivable – accounts payable)}$$

$$CF = NI - \text{change in net working capital (NWC)}$$

Operating Activities

Cash flow from operating activities	
Net income	3,160
Depreciation	3,000
Change in net working capital	<u>-1,000</u>
Cash flow from operating activities	<u>7,160</u>

$$3,160 + 3,000 - (-1,000) = 7,160$$

Investing Activities

Investing activities, cash spent buying fixed assets and cash received from selling them

Investing activities show cash flows from,

1. Capital purchased and sold

- **Capital**, a firm's fixed assets
- e.g., machinery, plant, land
- **Capital expenditure**, the amount spent on buying capital during the period
- Cash flow is *negative* if firms buy capital

Investing Activities

Cash flow from investing activities	
Capital expenditure	−3,000
Sales of fixed assets	200
Cash flow from investing activities	<u>−2,800</u>

$$-3,000 + 200 = -2,800$$

Financing Activities

Financing activities, cash raised from lenders and shareholders, and cash returned to them

Financing activities show cash flows from,

1. Selling new shares → Cash ↑
2. Paying dividends → Cash ↓
3. Taking new debt, either a loan or a bond → Cash ↑
4. Repaying principal amount of old debt → Cash ↓

Practice Problem

- During the year, a business sells new shares for \$120,000 and pays dividends of \$30,000. It takes a new loan of \$80,000 and repays \$50,000 of principal on old debt. How much is its cash flow from financing activities?

Practice Problem

- During the year, a business sells new shares for \$120,000 and pays dividends of \$30,000. It takes a new loan of \$80,000 and repays \$50,000 of principal on old debt. How much is its cash flow from financing activities?
- Cash flow from financing activities = $\$120,000 - \$30,000 + \$80,000 - \$50,000 = \$120,000$

Recap of Part 3

- The statement of cash flows, cash receipts and payments over a period
- Net income $\neq$ cash flow
- The statement has three parts
 - **Operating activities**, cash flow = Net income + Depreciation – Change in NWC
 - **Investing activities**, cash spent buying fixed assets and cash received from selling them
 - **Financing activities**, cash raised from lenders and shareholders, and cash returned to them

Practice Problem

- Which of the following will make cash flow higher than net income for a firm?
 - a. Increase in inventories.
 - b. Decrease in accounts payable.
 - c. Decrease in accounts receivable.
 - d. Expenditures on fixed assets.

Practice Problem

- Which of the following will make cash flow higher than net income for a firm?
 - a. Increase in inventories.
 - b. Decrease in accounts payable.
 - c. Decrease in accounts receivable.
 - d. Expenditures on fixed assets.

Practice Problem

- A business has net income of \$3,160 million and depreciation of \$3,000 million, and its NWC changed by $-\$1,000$ million. What is its cash flow from operating activities?

Practice Problem

- A business has net income of \$3,160 million and depreciation of \$3,000 million, and its NWC changed by $-\$1,000$ million. What is its cash flow from operating activities?
- Cash flow from operating activities = Net income + Depreciation – Change in NWC

Practice Problem

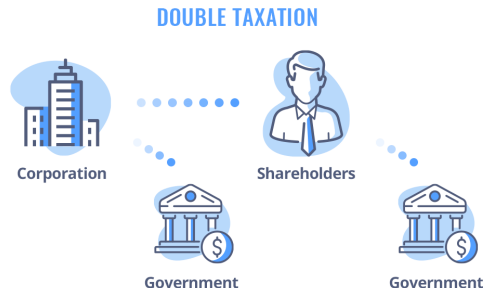
- A business has net income of \$3,160 million and depreciation of \$3,000 million, and its NWC changed by $-\$1,000$ million. What is its cash flow from operating activities?
- Cash flow from operating activities = Net income + Depreciation – Change in NWC
- $\$3,160 + \$3,000 - (-\$1,000) = \$7,160$ million

Module 2, Part 4

Taxation

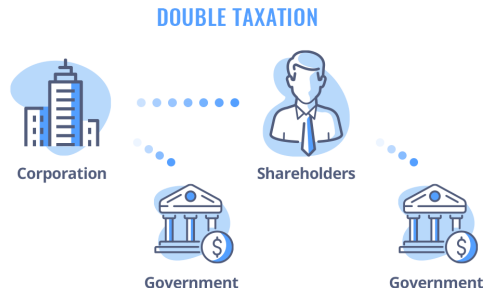
Corporate Taxation, Double Taxation

- Recall from **Module 1**,
 - The corporation pays tax on its income



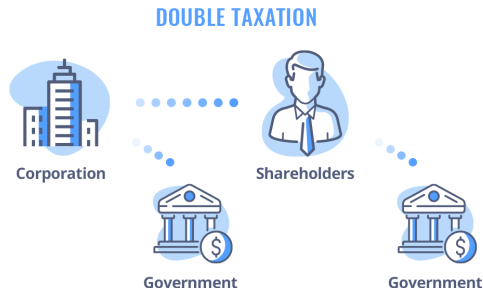
Corporate Taxation, Double Taxation

- Recall from **Module 1**,
 - The corporation pays tax on its income
 - Shareholders pay tax again on what they receive



Corporate Taxation, Double Taxation

- Recall from **Module 1**,
 - The corporation pays tax on its income
 - Shareholders pay tax again on what they receive
- This is **double taxation**



Corporate Taxation, The Tax Rate

- The U.S. federal corporate tax rate is 21%



WSJ, February 28, 2025 (6:57),
“Economists on How Trump’s 2017
Tax Cuts Actually Played Out”

Corporate Taxation, The Tax Rate

- The U.S. federal corporate tax rate is **21%**
- The rate is a major political topic



WSJ, February 28, 2025 (6:57),
“Economists on How Trump’s 2017
Tax Cuts Actually Played Out”

Corporate Taxation, The Tax Rate

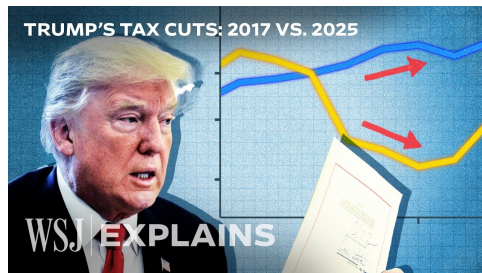
- The U.S. federal corporate tax rate is **21%**
- The rate is a major political topic
- **TCJA (2017)**, cut the rate from 35% to 21%



WSJ, February 28, 2025 (6:57),
“Economists on How Trump’s 2017
Tax Cuts Actually Played Out”

Corporate Taxation, The Tax Rate

- The U.S. federal corporate tax rate is **21%**
- The rate is a major political topic
- **TCJA (2017)**, cut the rate from 35% to 21%
- The tax cut was extended and largely made permanent in 2025



WSJ, February 28, 2025 (6:57),
“Economists on How Trump’s 2017
Tax Cuts Actually Played Out”

Tax Shields

- Depreciation and interest come first

Line item	Letters	Remember it as
Revenues	R	R ainbow
COGS	C	C andy
SG&A expenses	S	S ells
Depreciation	D	D ependably
EBIT	E	E specially
Interest expense	I	I n
Taxable income	T.I.	T exas I ncorporated
Taxes	T	T exas
Net income	N.I.	N ever I ncorporates
Dividends	D	D uring
Addition to retained earnings	R.E.	R e- E lections

Tax Shields

- Depreciation and interest come first
- Taxes come last

Line item	Letters	Remember it as
Revenues	R	R ainbow
COGS	C	C andy
SG&A expenses	S	S ells
Depreciation	D	D ependably
EBIT	E	E specially
Interest expense	I	I n
Taxable income	T.I.	T exas I ncorporated
Taxes	T	T exas
Net income	N.I.	N ever I ncorporates
Dividends	D	D uring
Addition to retained earnings	R.E.	R e- E lections

Tax Shields

- Depreciation and interest come first
- Taxes come last
- Depreciation and interest reduce corporate taxes

Line item	Letters	Remember it as
Revenues	R	R ainbow
COGS	C	C andy
SG&A expenses	S	S ells
Depreciation	D	D ependably
EBIT	E	E specially
Interest expense	I	I n
Taxable income	T.I.	T exas I ncorporated
Taxes	T	T exas
Net income	N.I.	N ever I ncorporates
Dividends	D	D uring
Addition to retained earnings	R.E.	R e- E lections

Interest Tax Shield

- Interest expense can
 - act as a “shield” against corporate taxation

Interest Tax Shield

- Interest expense can
 - act as a “shield” against corporate taxation
 - increase the *total* cash flow to investors (debt and equity combined)

Interest Tax Shield

Firm A	
EBIT	\$5,000
Interest expense	\$1,000
Taxable income	<u>?</u>
Taxes (tax rate = 21%)	<u>?</u>
Net income	<u>?</u>

Firm B	
EBIT	\$5,000
Interest expense	\$0
Taxable income	<u>?</u>
Taxes (tax rate = 21%)	<u>?</u>
Net income	<u>?</u>

Interest Tax Shield

Firm A	
EBIT	\$5,000
Interest expense	\$1,000
Taxable income	\$4,000
Taxes (tax rate = 21%)	\$840
Net income	\$3,160

Firm B	
EBIT	\$5,000
Interest expense	\$0
Taxable income	?
Taxes (tax rate = 21%)	?
Net income	?

Interest Tax Shield

Firm A	
EBIT	\$5,000
Interest expense	\$1,000
Taxable income	\$4,000
Taxes (tax rate = 21%)	\$840
Net income	\$3,160

Firm B	
EBIT	\$5,000
Interest expense	\$0
Taxable income	\$5,000
Taxes (tax rate = 21%)	\$1,050
Net income	\$3,950

Interest Tax Shield

Firm A

EBIT	\$5,000	
Interest expense	\$1,000	→ debt investors
Taxable income	\$4,000	
Taxes (tax rate = 21%)	\$840	
Net income	\$3,160	

Firm B

EBIT	\$5,000	
Interest expense	\$0	→ debt investors
Taxable income	\$5,000	
Taxes (tax rate = 21%)	\$1,050	
Net income	\$3,950	

Interest Tax Shield

Firm A

EBIT	\$5,000	
Interest expense	\$1,000	→ debt investors
Taxable income	\$4,000	
Taxes (tax rate = 21%)	\$840	
Net income	\$3,160	→ equity investors

Firm B

EBIT	\$5,000	
Interest expense	\$0	→ debt investors
Taxable income	\$5,000	
Taxes (tax rate = 21%)	\$1,050	
Net income	\$3,950	→ equity investors

Interest Tax Shield

Firm A

EBIT	\$5,000		
Interest expense	\$1,000	→ debt investors	} \$4,160
Taxable income	\$4,000		
Taxes (tax rate = 21%)	\$840		
Net income	\$3,160	→ equity investors	

Firm B

EBIT	\$5,000		
Interest expense	\$0	→ debt investors	} \$3,950
Taxable income	\$5,000		
Taxes (tax rate = 21%)	\$1,050		
Net income	\$3,950	→ equity investors	

Interest Tax Shield

Firm A

EBIT	\$5,000		
Interest expense	\$1,000	→ debt investors	} \$4,160
Taxable income	\$4,000		
Taxes (tax rate = 21%)	\$840		
Net income	\$3,160	→ equity investors	

Firm B

EBIT	\$5,000		
Interest expense	\$0	→ debt investors	} \$3,950
Taxable income	\$5,000		
Taxes (tax rate = 21%)	\$1,050		
Net income	\$3,950	→ equity investors	

Investors get \$4,160 in total with debt, \$3,950 without

Interest Tax Shield

- Interest expense can
 - act as a “shield” against corporate taxation
 - increase the *total* cash flow to investors (debt and equity combined)
- **Interest tax shield**, the tax saving that interest expense creates, as interest is deducted (subtracted) before taxes

Interest Tax Shield

- Interest expense can
 - act as a “shield” against corporate taxation
 - increase the *total* cash flow to investors (debt and equity combined)
- **Interest tax shield**, the tax saving that interest expense creates, as interest is deducted (subtracted) before taxes

$$\text{Interest tax shield} = \text{Tax rate} \times \text{Interest expense}$$

Interest Tax Shield

- Interest expense can
 - act as a “shield” against corporate taxation
 - increase the *total* cash flow to investors (debt and equity combined)
- **Interest tax shield**, the tax saving that interest expense creates, as interest is deducted (subtracted) before taxes

$$\text{Interest tax shield} = \text{Tax rate} \times \text{Interest expense}$$

- In the example, $21\% \times \$1,000 = \210

Interest Tax Shield

- Each dollar of interest lowers taxable income by \$1

Interest Tax Shield

- Each dollar of interest lowers taxable income by \$1
- So taxes fall by the tax rate $\times$ \$1

Interest Tax Shield

- Each dollar of interest lowers taxable income by \$1
- So taxes fall by the tax rate $\times$ \$1

$$\underbrace{\text{Net income}}_{\downarrow \text{ by } (1-\text{tax rate}) \times \$1} = \underbrace{\text{Taxable income}}_{\downarrow \text{ by } \$1} - \underbrace{\text{Taxes}}_{\uparrow \text{ by tax rate} \times \$1}$$

Interest Tax Shield

- Each dollar of interest lowers taxable income by \$1
- So taxes fall by the tax rate $\times$ \$1

$$\underbrace{\text{Net income}}_{\downarrow \text{ by } (1-\text{tax rate}) \times \$1} = \underbrace{\text{Taxable income}}_{\downarrow \text{ by } \$1} - \underbrace{\text{Taxes}}_{\uparrow \text{ by tax rate} \times \$1}$$

- Equity investors get $(1 - \text{tax rate}) \times \1 less

Interest Tax Shield

- Each dollar of interest lowers taxable income by \$1
- So taxes fall by the tax rate $\times$ \$1

$$\underbrace{\text{Net income}}_{\downarrow \text{ by } (1-\text{tax rate}) \times \$1} = \underbrace{\text{Taxable income}}_{\downarrow \text{ by } \$1} - \underbrace{\text{Taxes}}_{\uparrow \text{ by tax rate} \times \$1}$$

- Equity investors get $(1 - \text{tax rate}) \times \1 less
- Debt investors get \$1 more

Interest Tax Shield

- Each dollar of interest lowers taxable income by \$1
- So taxes fall by the tax rate $\times$ \$1

$$\underbrace{\text{Net income}}_{\downarrow \text{ by } (1-\text{tax rate}) \times \$1} = \underbrace{\text{Taxable income}}_{\downarrow \text{ by } \$1} - \underbrace{\text{Taxes}}_{\uparrow \text{ by tax rate} \times \$1}$$

- Equity investors get $(1 - \text{tax rate}) \times \1 less
- Debt investors get \$1 more

$$\underbrace{\text{Total CF to investors}}_{\uparrow \text{ by tax rate} \times \$1} = \underbrace{\text{Debt investors}}_{\uparrow \text{ by } \$1} + \underbrace{\text{Equity investors}}_{\downarrow \text{ by } (1-\text{tax rate}) \times \$1}$$

Interest Tax Shield

- Each dollar of interest lowers taxable income by \$1
- So taxes fall by the tax rate $\times$ \$1

$$\underbrace{\text{Net income}}_{\downarrow \text{ by } (1-\text{tax rate})\times \$1} = \underbrace{\text{Taxable income}}_{\downarrow \text{ by } \$1} - \underbrace{\text{Taxes}}_{\uparrow \text{ by tax rate}\times \$1}$$

- Equity investors get $(1 - \text{tax rate}) \times \1 less
- Debt investors get \$1 more

$$\underbrace{\text{Total CF to investors}}_{\uparrow \text{ by tax rate}\times \$1} = \underbrace{\text{Debt investors}}_{\uparrow \text{ by } \$1} + \underbrace{\text{Equity investors}}_{\downarrow \text{ by } (1-\text{tax rate})\times \$1}$$

- Each dollar of interest raises the *total* cash flow to investors by the tax rate $\times$ \$1

Interest Tax Shield

- Each dollar of interest lowers taxable income by \$1
- So taxes fall by the tax rate $\times$ \$1

$$\underbrace{\text{Net income}}_{\downarrow \text{ by } (1-\text{tax rate}) \times \$1} = \underbrace{\text{Taxable income}}_{\downarrow \text{ by } \$1} - \underbrace{\text{Taxes}}_{\uparrow \text{ by tax rate} \times \$1}$$

- Equity investors get $(1 - \text{tax rate}) \times \1 less
- Debt investors get \$1 more

$$\underbrace{\text{Total CF to investors}}_{\uparrow \text{ by tax rate} \times \$1} = \underbrace{\text{Debt investors}}_{\uparrow \text{ by } \$1} + \underbrace{\text{Equity investors}}_{\downarrow \text{ by } (1-\text{tax rate}) \times \$1}$$

- Each dollar of interest raises the *total* cash flow to investors by the tax rate $\times$ \$1
- Should a firm take on as much debt as possible? No, bankruptcy costs.

Depreciation Tax Shield

- Depreciation can also
 - act as a “shield” against corporate taxation

Depreciation Tax Shield

- Depreciation can also
 - act as a “shield” against corporate taxation
 - increase the *total* cash flow to investors (debt and equity combined)

Depreciation Tax Shield

- Example, Suppose a business has an *additional* \$1,000 million in depreciation, and its tax rate is 21%

Income Statement (\$ Million)	
	Baseline
Revenues	80,000
COGS	56,000
SG&A expenses	16,000
Depreciation	3,000
EBIT	5,000
Interest expense	1,000
Taxable income	4,000
Taxes	840
Net income	3,160

Depreciation Tax Shield

- Example, Suppose a business has an *additional* \$1,000 million in depreciation, and its tax rate is 21%

Income Statement (\$ Million)		
	Baseline	More depreciation
Revenues	80,000	80,000
COGS	56,000	56,000
SG&A expenses	16,000	16,000
Depreciation	3,000	
EBIT	5,000	
Interest expense	1,000	
Taxable income	4,000	
Taxes	840	
Net income	3,160	

Depreciation Tax Shield

- Example, Suppose a business has an *additional* \$1,000 million in depreciation, and its tax rate is 21%

Income Statement (\$ Million)		
	Baseline	More depreciation
Revenues	80,000	80,000
COGS	56,000	56,000
SG&A expenses	16,000	16,000
Depreciation	3,000	4,000
EBIT	5,000	
Interest expense	1,000	
Taxable income	4,000	
Taxes	840	
Net income	3,160	

Depreciation Tax Shield

- Example, Suppose a business has an *additional* \$1,000 million in depreciation, and its tax rate is 21%

Income Statement (\$ Million)		
	Baseline	More depreciation
Revenues	80,000	80,000
COGS	56,000	56,000
SG&A expenses	16,000	16,000
Depreciation	3,000	4,000
EBIT	5,000	4,000
Interest expense	1,000	
Taxable income	4,000	
Taxes	840	
Net income	3,160	

Depreciation Tax Shield

- Example, Suppose a business has an *additional* \$1,000 million in depreciation, and its tax rate is 21%

Income Statement (\$ Million)		
	Baseline	More depreciation
Revenues	80,000	80,000
COGS	56,000	56,000
SG&A expenses	16,000	16,000
Depreciation	3,000	4,000
EBIT	5,000	4,000
Interest expense	1,000	1,000
Taxable income	4,000	
Taxes	840	
Net income	3,160	

Depreciation Tax Shield

- Example, Suppose a business has an *additional* \$1,000 million in depreciation, and its tax rate is 21%

Income Statement (\$ Million)		
	Baseline	More depreciation
Revenues	80,000	80,000
COGS	56,000	56,000
SG&A expenses	16,000	16,000
Depreciation	3,000	4,000
EBIT	5,000	4,000
Interest expense	1,000	1,000
Taxable income	4,000	3,000
Taxes	840	
Net income	3,160	

Depreciation Tax Shield

- Example, Suppose a business has an *additional* \$1,000 million in depreciation, and its tax rate is 21%

Income Statement (\$ Million)		
	Baseline	More depreciation
Revenues	80,000	80,000
COGS	56,000	56,000
SG&A expenses	16,000	16,000
Depreciation	3,000	4,000
EBIT	5,000	4,000
Interest expense	1,000	1,000
Taxable income	4,000	3,000
Taxes	840	630
Net income	3,160	

Depreciation Tax Shield

- Example, Suppose a business has an *additional* \$1,000 million in depreciation, and its tax rate is 21%

Income Statement (\$ Million)		
	Baseline	More depreciation
Revenues	80,000	80,000
COGS	56,000	56,000
SG&A expenses	16,000	16,000
Depreciation	3,000	4,000
EBIT	5,000	4,000
Interest expense	1,000	1,000
Taxable income	4,000	3,000
Taxes	840	630
Net income	3,160	2,370

Depreciation Tax Shield

- Net income *falls* by \$790 million

\$ Million	Baseline	More depreciation	Change
Net income	3,160	2,370	−790
Add back depreciation	_____	_____	_____
Cash flow	_____	_____	_____

Depreciation Tax Shield

- Net income *falls* by \$790 million
- Recall Part 3, Cash flow from operating activities = Net income + Depreciation – Change in NWC

\$ Million	Baseline	More depreciation	Change
Net income	3,160	2,370	–790
Add back depreciation	3,000	4,000	+1,000
Cash flow			

Depreciation Tax Shield

- Net income *falls* by \$790 million
- Recall Part 3, Cash flow from operating activities = Net income + Depreciation – Change in NWC
- Cash flow *rises* by \$210 million

\$ Million	Baseline	More depreciation	Change
Net income	3,160	2,370	–790
Add back depreciation	3,000	4,000	+1,000
Cash flow	6,160	6,370	+210

Depreciation Tax Shield

- Depreciation can also
 - act as a “shield” against corporate taxation
 - increase the *total* cash flow to investors (debt and equity combined)
- Depreciation reduces taxable income, cuts taxes, and raises cash flow

Depreciation Tax Shield

- Depreciation can also
 - act as a “shield” against corporate taxation
 - increase the *total* cash flow to investors (debt and equity combined)
- Depreciation reduces taxable income, cuts taxes, and raises cash flow
- Each dollar of depreciation raises the *total* cash flow to investors by the tax rate $\times \$1$

Depreciation Tax Shield

- Depreciation can also
 - act as a “shield” against corporate taxation
 - increase the *total* cash flow to investors (debt and equity combined)
- Depreciation reduces taxable income, cuts taxes, and raises cash flow
- Each dollar of depreciation raises the *total* cash flow to investors by the tax rate $\times \$1$

$$\text{Depreciation tax shield} = \text{Tax rate} \times \text{Depreciation}$$

Depreciation Tax Shield

- Each dollar of depreciation lowers taxable income by \$1

Depreciation Tax Shield

- Each dollar of depreciation lowers taxable income by \$1
- So taxes fall by the tax rate $\times$ \$1

Depreciation Tax Shield

- Each dollar of depreciation lowers taxable income by \$1
- So taxes fall by the tax rate $\times$ \$1

$$\underbrace{\text{Net income}}_{\downarrow \text{ by } (1-\text{tax rate}) \times \$1} = \underbrace{\text{Taxable income}}_{\downarrow \text{ by } \$1} - \underbrace{\text{Taxes}}_{\uparrow \text{ by tax rate} \times \$1}$$

Depreciation Tax Shield

- Each dollar of depreciation lowers taxable income by \$1
- So taxes fall by the tax rate $\times$ \$1

$$\underbrace{\text{Net income}}_{\downarrow \text{ by } (1-\text{tax rate}) \times \$1} = \underbrace{\text{Taxable income}}_{\downarrow \text{ by } \$1} - \underbrace{\text{Taxes}}_{\uparrow \text{ by tax rate} \times \$1}$$

$$\underbrace{\text{CF from operating activities}}_{\uparrow \text{ by tax rate} \times \$1} = \underbrace{\text{Net income}}_{\downarrow \text{ by } (1-\text{tax rate}) \times \$1} + \underbrace{\text{Depreciation}}_{\uparrow \text{ by } \$1} - \underbrace{\text{Change in NWC}}_{\text{No change}}$$

Depreciation Tax Shield

- Each dollar of depreciation lowers taxable income by \$1
- So taxes fall by the tax rate $\times$ \$1

$$\underbrace{\text{Net income}}_{\downarrow \text{ by } (1-\text{tax rate}) \times \$1} = \underbrace{\text{Taxable income}}_{\downarrow \text{ by } \$1} - \underbrace{\text{Taxes}}_{\uparrow \text{ by tax rate} \times \$1}$$

$$\underbrace{\text{CF from operating activities}}_{\uparrow \text{ by tax rate} \times \$1} = \underbrace{\text{Net income}}_{\downarrow \text{ by } (1-\text{tax rate}) \times \$1} + \underbrace{\text{Depreciation}}_{\uparrow \text{ by } \$1} - \underbrace{\text{Change in NWC}}_{\text{No change}}$$

- Equity investors get tax rate $\times$ \$1 more

Depreciation Tax Shield

- Each dollar of depreciation lowers taxable income by \$1
- So taxes fall by the tax rate $\times$ \$1

$$\underbrace{\text{Net income}}_{\downarrow \text{ by } (1-\text{tax rate}) \times \$1} = \underbrace{\text{Taxable income}}_{\downarrow \text{ by } \$1} - \underbrace{\text{Taxes}}_{\uparrow \text{ by tax rate} \times \$1}$$

$$\underbrace{\text{CF from operating activities}}_{\uparrow \text{ by tax rate} \times \$1} = \underbrace{\text{Net income}}_{\downarrow \text{ by } (1-\text{tax rate}) \times \$1} + \underbrace{\text{Depreciation}}_{\uparrow \text{ by } \$1} - \underbrace{\text{Change in NWC}}_{\text{No change}}$$

- Equity investors get tax rate $\times$ \$1 more
- Each dollar of depreciation raises the *total* cash flow to investors by the tax rate $\times$ \$1

Depreciation Tax Shield

- Each dollar of depreciation lowers taxable income by \$1
- So taxes fall by the tax rate $\times$ \$1

$$\underbrace{\text{Net income}}_{\downarrow \text{ by } (1-\text{tax rate})\times \$1} = \underbrace{\text{Taxable income}}_{\downarrow \text{ by } \$1} - \underbrace{\text{Taxes}}_{\uparrow \text{ by tax rate}\times \$1}$$

$$\underbrace{\text{CF from operating activities}}_{\uparrow \text{ by tax rate}\times \$1} = \underbrace{\text{Net income}}_{\downarrow \text{ by } (1-\text{tax rate})\times \$1} + \underbrace{\text{Depreciation}}_{\uparrow \text{ by } \$1} - \underbrace{\text{Change in NWC}}_{\text{No change}}$$

- Equity investors get tax rate $\times$ \$1 more
- Each dollar of depreciation raises the *total* cash flow to investors by the tax rate $\times$ \$1
- Can a firm have as much depreciation as possible? No, constrained by cost of acquisition and accounting rules.

Practice Problem

- A business pays a 21% tax rate. Suppose it had an additional \$200 million in depreciation. By how much would its taxes fall? What about its cash flow?

Practice Problem

- A business pays a 21% tax rate. Suppose it had an additional \$200 million in depreciation. By how much would its taxes fall? What about its cash flow?
- Taxes fall by $21\% \times \$200 \text{ million} = \42 million

Practice Problem

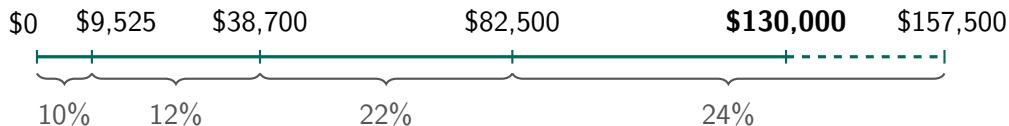
- A business pays a 21% tax rate. Suppose it had an additional \$200 million in depreciation. By how much would its taxes fall? What about its cash flow?
- Taxes fall by $21\% \times \$200 \text{ million} = \42 million
- Cash flow rises by the same \$42 million, the depreciation tax shield

Personal Taxation

- Personal tax rates matter for salaried employees and non-corporate businesses
- Marginal tax rate = Tax rate that is applied to each extra dollar of income
- Average tax rate = $\text{Total taxes} / \text{Total income}$

Personal Taxation

Taxable Income (Single Taxpayers)	Marginal Tax Rate (%)
\$0–\$9,525	10
\$9,526–\$38,700	12
\$38,701–\$82,500	22
\$82,501–\$157,500	24
\$157,501–\$200,000	32
\$200,001–\$500,000	35
\$500,001 and above	37



Example, Average Tax Rate

Suppose your salary is \$130,000 per year. What would be your average tax rate?

Taxable Income (Single Taxpayers)	Marginal Tax Rate (%)
\$0–\$9,525	10
\$9,526–\$38,700	12
\$38,701–\$82,500	22
\$82,501–\$157,500	24
\$157,501–\$200,000	32
\$200,001–\$500,000	35
\$500,001 and above	37

- Total tax = $\$9,525 \times 10\% + \$ (38,700 - 9,525) \times 12\% + \$ (82,500 - 38,700) \times 22\% + \$ (130,000 - 82,500) \times 24\% = \$25,489.5$
- Average tax rate = $25,489.5 / 130,000 = 19.61\%$

Practice Problem

- Suppose your salary is \$30,000 per year. What would be your average tax rate? What would be your marginal tax rate?

Taxable Income (Single Taxpayers)	Marginal Tax Rate (%)
\$0–\$9,525	10
\$9,526–\$38,700	12
\$38,701–\$82,500	22
\$82,501–\$157,500	24
\$157,501–\$200,000	32
\$200,001–\$500,000	35
\$500,001 and above	37

Practice Problem

- Suppose your salary is \$30,000 per year. What would be your average tax rate? What would be your marginal tax rate?

Taxable Income (Single Taxpayers)	Marginal Tax Rate (%)
\$0–\$9,525	10
\$9,526–\$38,700	12
\$38,701–\$82,500	22
\$82,501–\$157,500	24
\$157,501–\$200,000	32
\$200,001–\$500,000	35
\$500,001 and above	37

- Total tax = $\$9,525 \times 10\% + (\$30,000 - \$9,525) \times 12\% = \$3,409.5$

Practice Problem

- Suppose your salary is \$30,000 per year. What would be your average tax rate? What would be your marginal tax rate?

Taxable Income (Single Taxpayers)	Marginal Tax Rate (%)
\$0–\$9,525	10
\$9,526–\$38,700	12
\$38,701–\$82,500	22
\$82,501–\$157,500	24
\$157,501–\$200,000	32
\$200,001–\$500,000	35
\$500,001 and above	37

- Total tax = $\$9,525 \times 10\% + (\$30,000 - \$9,525) \times 12\% = \$3,409.5$
- Average tax rate = $3,409.5 / 30,000 = 11.37\%$

Practice Problem

- Suppose your salary is \$30,000 per year. What would be your average tax rate? What would be your marginal tax rate?

Taxable Income (Single Taxpayers)	Marginal Tax Rate (%)
\$0–\$9,525	10
\$9,526–\$38,700	12
\$38,701–\$82,500	22
\$82,501–\$157,500	24
\$157,501–\$200,000	32
\$200,001–\$500,000	35
\$500,001 and above	37

- Total tax = $\$9,525 \times 10\% + (\$30,000 - \$9,525) \times 12\% = \$3,409.5$
- Average tax rate = $3,409.5 / 30,000 = 11.37\%$
- Marginal tax rate = 12%

Recap of Part 4

- The U.S. federal corporate tax rate is **21%**
- Interest and depreciation come before taxes, so both shield income from tax
- A tax shield is worth the tax rate times the deduction, leaving more cash for investors
- Marginal rate applies to each extra dollar of income, average rate is total taxes over total income

Key Takeaways

- Three statements, what a firm owns and owes, what it earned, and how cash moved

Key Takeaways

- Three statements, what a firm owns and owes, what it earned, and how cash moved
- Book values are the cost of acquiring real assets or the amount raised from selling financial assets, market values are what things would sell for today

Key Takeaways

- Three statements, what a firm owns and owes, what it earned, and how cash moved
- Book values are the cost of acquiring real assets or the amount raised from selling financial assets, market values are what things would sell for today
- Net income $\neq$ cash flow, add back depreciation, subtract the change in NWC

Key Takeaways

- Three statements, what a firm owns and owes, what it earned, and how cash moved
- Book values are the cost of acquiring real assets or the amount raised from selling financial assets, market values are what things would sell for today
- Net income $\neq$ cash flow, add back depreciation, subtract the change in NWC
- Interest and depreciation shield taxes, leaving more cash for investors

Practice Problem

- A business pays a 21% tax rate. Its COGS is \$20,000, its SG&A expenses are \$10,000, and its depreciation is \$8,000. Its interest expense is \$2,000, and its taxes are \$2,100. How much is its net income? How much is its EBIT? How much are its revenues?

Practice Problem

- A business pays a 21% tax rate. Its COGS is \$20,000, its SG&A expenses are \$10,000, and its depreciation is \$8,000. Its interest expense is \$2,000, and its taxes are \$2,100. How much is its net income? How much is its EBIT? How much are its revenues?
- $\text{Taxable income} = \text{Taxes} / \text{Tax rate} = \$2,100 / 21\% = \$10,000$

Practice Problem

- A business pays a 21% tax rate. Its COGS is \$20,000, its SG&A expenses are \$10,000, and its depreciation is \$8,000. Its interest expense is \$2,000, and its taxes are \$2,100. How much is its net income? How much is its EBIT? How much are its revenues?
- Taxable income = Taxes / Tax rate = $\$2,100 / 21\% = \$10,000$
- Net income = Taxable income – Taxes = $\$10,000 - \$2,100 = \$7,900$

Practice Problem

- A business pays a 21% tax rate. Its COGS is \$20,000, its SG&A expenses are \$10,000, and its depreciation is \$8,000. Its interest expense is \$2,000, and its taxes are \$2,100. How much is its net income? How much is its EBIT? How much are its revenues?
- Taxable income = Taxes / Tax rate = $\$2,100 / 21\% = \$10,000$
- Net income = Taxable income – Taxes = $\$10,000 - \$2,100 = \$7,900$
- EBIT = Taxable income + Interest expense = $\$10,000 + \$2,000 = \$12,000$

Practice Problem

- A business pays a 21% tax rate. Its COGS is \$20,000, its SG&A expenses are \$10,000, and its depreciation is \$8,000. Its interest expense is \$2,000, and its taxes are \$2,100. How much is its net income? How much is its EBIT? How much are its revenues?
- Taxable income = Taxes / Tax rate = $\$2,100 / 21\% = \$10,000$
- Net income = Taxable income – Taxes = $\$10,000 - \$2,100 = \$7,900$
- EBIT = Taxable income + Interest expense = $\$10,000 + \$2,000 = \$12,000$
- Revenues = EBIT + COGS + SG&A expenses + Depreciation = $\$12,000 + \$20,000 + \$10,000 + \$8,000 = \$50,000$

Practice Problem

If a firm's net income is positive and its depreciation is positive, which of the following could account for a negative cash flow from operating activities?

- a. Inventories decrease while accounts receivable and accounts payable stay the same.
- b. Accounts receivable increases while inventories and accounts payable stay the same.
- c. The firm raises money by issuing stocks.
- d. A large addition is made to plant and equipment.

Practice Problem

If a firm's net income is positive and its depreciation is positive, which of the following could account for a negative cash flow from operating activities?

- a. Inventories decrease while accounts receivable and accounts payable stay the same.
- b. Accounts receivable increases while inventories and accounts payable stay the same.
- c. The firm raises money by issuing stocks.
- d. A large addition is made to plant and equipment.

Thank you!

Please let me know if you have any questions.